PMBOK® GUIDE 8TH EDITION | PROCESSES FLOW

RICARDO VARGAS

VALUE-DRIVEN

FOCUS ON VALUE

INTEGRATE SUSTAINABILITY

EMBED QUALITY

BE AN ACCOUNTABLE LEADER

PROACTIVE

OWNERSHIP

ADOPT A HOLISTIC VIEW

BUILD AN EMPOWERED CULTURE

INITIATING

1.1 GOVERNANCE — Initiate Project or Phase

INPUTS
- Business documents
  - Business case
  - Benefits management plan
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Brainstorming
  - Focus groups
  - Interviews
- Interpersonal and team skills
  - Conflict management
  - Facilitation
  - Meeting management
- Meetings
- Responsibility assignment matrix
- Project canvas
- Etc.

OUTPUTS
- Project charter
- Assumption log
- Etc.

5.1 STAKEHOLDERS — Identify Stakeholders

INPUTS
- Project charter
- Business documents
  - Business case
  - Benefits management plan
- Project management plan
  - Communications management plan
  - Stakeholder engagement plan
- Project documents
  - Change log
  - Issue log
  - Requirements documentation
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Questionnaires and surveys
  - Brainstorming
- Data analysis
  - Stakeholder analysis
  - Document analysis
- Data representation
  - Stakeholder mapping/—representation
- Meetings
- Etc.

OUTPUTS
- Stakeholder register
- Change requests
- Project management plan updates
  - Requirements management plan
  - Communications management plan
  - Risk management plan
  - Stakeholder engagement plan
- Project document updates
  - Assumption log
  - Issue log
  - Risk register
- Etc.

PLANNING

2.1 SCOPE — Plan Scope Management

INPUTS
- Project charter
- Project management plan
- Project documents
  - Requirements documentation
  - Risk register
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Interviews
  - Focus groups
  - Questionnaires and surveys
- Data analysis
- Test and inspection planning
- Etc.

OUTPUTS
- Project management plan updates
  - Scope management plan
  - Requirements management plan
- Etc.

2.2 SCOPE — Elicit and Analyze Requirements

INPUTS
- Project charter
- Agreements
- Business case
- Project documents
  - Assumption log
  - Lessons learned register
  - Stakeholder register
- Project management plan
  - Requirements management plan
  - Scope management plan
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Decision-making
- Data gathering
  - Benchmarking
  - Brainstorming
  - Focus groups
  - Interviews
  - Questionnaires and surveys
- Data analysis
  - Document analysis
- Data representation
- Interpersonal and team skills
  - Nominal group
- Design thinking
- Prioritization ranking
- Meetings
- Etc.

OUTPUTS
- Requirements documentation
- Etc.

2.3 SCOPE — Define Scope

INPUTS
- Project charter
- Assumption log
- Project management plan
- Requirements documentation
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Decision-making
- Data analysis
- Decomposition
- Interpersonal and team skills
  - Facilitation
- Product analysis
- Etc.

OUTPUTS
- Project documents
  - Project scope statement
  - Requirements documentation
- Etc.

2.4 SCOPE — Develop Scope Structure

INPUTS
- Project management plan
- Project documents
  - Project scope statement
  - Requirements documentation
- Approved changes
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Brainstorming
- Decomposition
- Etc.

OUTPUTS
- Scope baseline
- Work breakdown structure (WBS)
- WBS dictionary
- User stories
- Product backlog
- Etc.

1.2 GOVERNANCE — Integrate and Align Project Plans

INPUTS
- Project charter
- Outputs from other processes
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Brainstorming
  - Checklists
  - Focus groups
  - Interviews
- Interpersonal and team skills
  - Conflict management
  - Facilitation
  - Meeting management
- Meetings
- Project canvas
- Etc.

OUTPUTS
- Project management plan
- Etc.

1.3 GOVERNANCE — Plan Sourcing Strategy

INPUTS
- Project charter
- Project management plan
  - Scope management plan
  - Quality management plan
  - Scope baseline
  - Schedule management plan
  - Financial management plan
  - Resource management plan
- Project documents
  - Milestone list
  - Requirements documentation
  - Requirements traceability matrix
  - Quality metrics
  - Resource requirements
  - Project team assignments
  - Risk register
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Market research
- Make-or-buy analysis
- Source selection analysis
- Document analysis
- Etc.

OUTPUTS
- Sourcing strategy plan
  - Insourcing or outsourcing decisions
  - Source selection criteria
- Etc.

4.1 FINANCE — Plan Financial Management

INPUTS
- Project charter
- Project management plan
  - Schedule management plan
  - Risk management plan
- Project documents
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data analysis
  - Alternative analysis
- Meetings
- Etc.

OUTPUTS
- Financial management plan
- Funding strategy
- Etc.

4.2 FINANCE — Estimate Costs

INPUTS
- Project management plan
  - Quality management plan
  - Scope baseline
- Project documents
  - Lessons learned register
  - Project schedule
  - Resource requirements
  - Risk register
- Make-or-buy decisions
- Work package estimation
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Analogous estimating
- Parametric estimating
- Bottom-up estimating
- Multipoint estimating
- Data analysis
  - Alternative analysis
  - Reserve analysis
  - Cost of quality
- Project management information system
- Decision-making
  - Voting
- Etc.

OUTPUTS
- Cost estimates
- Basis of estimates
- Project document updates
  - Assumption log
  - Lessons learned register
  - Risk register
- Etc.

4.3 FINANCE — Develop Budget

INPUTS
- Project management plan
  - Financial management plan
  - Resource management plan
  - Scope baseline
- Project documents
  - Basis of estimates
  - Cost estimates
  - Project schedule
  - Risk register
- Business documents
  - Business case
  - Benefits management plan
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Cost aggregation
- Data analysis
  - Reserve analysis
- Historical information review
- Funding limit reconciliation
- Financing
- Etc.

OUTPUTS
- Cost baseline
- Project funding requirements
- Project document updates
  - Cost estimates
  - Project schedule
  - Risk register
- Etc.

6.1 RESOURCES — Plan Resource Management

INPUTS
- Project charter
- Project management plan
  - Quality management plan
  - Scope baseline
- Project documents
  - Project schedule
  - Requirements documentation
  - Risk register
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Interviews
- Data analysis
  - SWOT analysis
- Data representation
  - Hierarchical charts
  - Responsibility assignment matrix
  - Text-oriented formats
- Organizational theory
- Meetings
- Green human resource management
- Resource-based view
- Etc.

OUTPUTS
- Project management plan updates
  - Resource management plan
- Team charter
- Project document updates
  - Assumption log
  - Risk register
- Etc.

6.2 RESOURCES — Estimate Resources

INPUTS
- Project management plan
  - Resource management plan
  - Schedule management plan
  - Scope baseline
- Project documents
  - Activity attributes
  - Activity list
  - Assumption log
  - Cost estimates
  - Resource calendars
  - Risk register
- Project schedule
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Bottom-up estimating
- Analogous estimating
- Parametric estimating
- Data analysis
  - Alternative analysis
- Project management information system
- Meetings
- Data gathering
  - Interviews
- Artificial intelligence
- Predictive analytics
- Virtual reality
- Augmented reality
- Branch and bound
- Genetic algorithms
- Constructive cost model
- Etc.

OUTPUTS
- Resource requirements
- Basis of estimates
- Resource breakdown structure
- Project document updates
  - Activity attributes
  - Assumption log
  - Lessons learned register
  - Risk register
- Etc.

3.1 SCHEDULE — Plan Schedule Management

INPUTS
- Project charter
- Project management plan
  - Scope management plan
- Development approach
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data analysis
  - Alternative analysis
- Meetings
- Etc.

OUTPUTS
- Project management plan updates
  - Schedule management plan
- Etc.

3.2 SCHEDULE — Develop Schedule

INPUTS
- Project charter
- Project management plan
  - Scope management plan
- Development approach
- Project documents
  - Activity attributes
  - Activity list
  - Assumption log
  - Basis of estimates
  - Duration estimates
  - Lessons learned register
  - Milestone list
  - Project schedule network diagrams
  - Project team assignments
  - Resource calendars
  - Resource requirements
  - Risk register
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Decomposition
- Rolling wave planning
- Precedence diagramming method
- Logical relationship
- Leads and lags
- Dependency determination and integration
- Estimation techniques
- Reserve analysis
- Data analysis
  - What-if analysis
  - Simulation
  - Alternative analysis
- Voting
- Schedule network analysis
- Schedule compression
- Critical path method
- Critical chain method
- Resource optimization
  - Resource leveling
- Project management information system
- Agile release planning
- Etc.

OUTPUTS
- Schedule baseline
- Project schedule
- Schedule data
- Project calendars
- Change requests
- Project management plan updates
  - Schedule management plan
- Project document updates
  - Activity list
  - Activity attributes
  - Assumption log
  - Basis of estimates
  - Duration estimates
  - Lessons learned register
  - Milestone list
  - Project schedule network diagrams
  - Resource requirements
  - Risk register
- Etc.

5.2 STAKEHOLDERS — Plan Stakeholder Engagement

INPUTS
- Project charter
- Project management plan
  - Resource management plan
  - Communications management plan
  - Risk management plan
- Project documents
  - Assumption log
  - Change log
  - Issue log
  - Project schedule
  - Risk register
  - Stakeholder register
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Benchmarking
- Data analysis
  - Assumption and constraint analysis
  - Root cause analysis
- Decision-making
  - Prioritization/—ranking
- Data representation
  - Mind mapping
  - Stakeholder engagement assessment matrix
- Meetings
- Etc.

OUTPUTS
- Project management plan updates
  - Stakeholder engagement plan
- Etc.

5.3 STAKEHOLDERS — Plan Communications Management

INPUTS
- Project charter
- Project management plan
  - Resource management plan
  - Stakeholder engagement plan
- Project documents
  - Requirements documentation
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Communication requirements analysis
- Communication technology
- Communication models
- Communication methods
- Interpersonal and team skills
  - Communication styles assessment
  - Political awareness
  - Cultural awareness
- Data representation
  - Stakeholder engagement assessment matrix
- Meetings
- Etc.

OUTPUTS
- Project management plan updates
  - Communications management plan
  - Stakeholder engagement plan
- Project document updates
  - Project schedule
  - Stakeholder register
- Etc.

7.1 RISK — Plan Risk Management

INPUTS
- Project charter
- Project management plan
  - All components
- Project documents
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Interviews
- Data analysis
  - Stakeholder analysis
- Meetings
- Etc.

OUTPUTS
- Project management plan updates
  - Risk management plan
- Etc.

7.2 RISK — Identify Risks

INPUTS
- Project management plan
  - Requirements management plan
  - Schedule management plan
  - Financial management plan
  - Quality management plan
  - Resource management plan
  - Risk management plan
  - Scope baseline
  - Schedule baseline
  - Cost baseline
- Project documents
  - Assumption log
  - Cost estimates
  - Duration estimates
  - Issue log
  - Lessons learned register
  - Requirements documentation
  - Resource requirements
  - Stakeholder register
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Brainstorming
  - Checklists
  - Interviews
- Data analysis
  - Root cause analysis
  - Assumption and constraint analysis
  - SWOT analysis
  - Document analysis
- Interpersonal and team skills
  - Facilitation
- Prompt lists
- Meetings
- Artificial intelligence
- Etc.

OUTPUTS
- Risk register
- Risk report
- Project document updates
  - Assumption log
  - Issue log
  - Lessons learned register
- Etc.

7.3 RISK — Perform Risk Analysis

INPUTS
- Project management plan
  - Risk management plan
  - Scope baseline
  - Schedule baseline
  - Cost baseline
- Project documents
  - Assumption log
  - Cost estimates
  - Duration estimates
  - Resource requirements
  - Risk register
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering and analysis
  - Interviews
- Interpersonal and team skills
  - Facilitation
- Risk categorization
- Data analysis
  - Risk probability and impact assessment
  - Simulations
  - Sensitivity analysis
  - Decision tree analysis
  - Influence diagrams
- Data representation
  - Probability and impact matrix
- Etc.

OUTPUTS
- Project document updates
  - Assumption log
  - Issue log
  - Risk register
  - Risk report
- Etc.

7.4 RISK — Plan Risk Responses

INPUTS
- Project management plan
  - Resource management plan
  - Risk management plan
  - Cost baseline
- Project documents
  - Lessons learned register
  - Project schedule
  - Project team assignments
  - Resource calendars
  - Risk register
  - Risk report
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data gathering
  - Interviews
- Interpersonal and team skills
  - Facilitation
- Strategies for threats
- Strategies for opportunities
- Contingent response strategies
- Strategies for overall project risk
- Data analysis
  - Alternative analysis
  - Cost-benefit analysis
- Decision-making
  - Multicriteria decision analysis
- Etc.

OUTPUTS
- Change requests
- Project management plan updates
  - Schedule management plan
  - Financial management plan
  - Quality management plan
  - Resource management plan
  - Procurement management plan
  - Scope baseline
  - Schedule baseline
  - Cost baseline
- Project document updates
  - Assumption log
  - Cost forecasts
  - Lessons learned register
  - Project schedule
  - Project team assignments
  - Risk register
  - Risk report
- Etc.

EXECUTING

6.3 RESOURCES — Acquire Resources

INPUTS
- Project management plan
  - Resource management plan
  - Procurement management plan
  - Cost baseline
- Project documents
  - Project schedule
  - Resource calendars
  - Resource requirements
  - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Decision-making
  - Multicriteria decision analysis
- Interpersonal and team skills
  - Negotiation
  - Problem-solving
- Preassignment
- Virtual teams
- Etc.

OUTPUTS
- Physical or virtual resource assignments
- Project team assignments
- Resource calendars
- Change requests
- Project management plan updates
  - Resource management plan
  - Cost baseline
- Project document updates
  - Lessons learned register
  - Project schedule
  - Resource breakdown structure
  - Resource requirements
  - Risk register
  - Stakeholder register
- Enterprise environmental factor updates
- Organizational process asset updates
- Etc.

6.4 RESOURCES — Lead the Team

INPUTS
- Project management plan
  - Resource management plan
- Project documents
  - Issue log
  - Lessons learned register
  - Project schedule
  - Resource calendars
  - Team charter
  - Work performance reports
  - Project team assignments
- Team performance assessments
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Colocation
- Virtual teams
- Communication technology
- Interpersonal and team skills
  - Conflict management
  - Influencing
  - Motivation
  - Negotiation
  - Team building
  - Decision-making
  - Critical thinking
  - Coaching and mentoring
  - Training
- Problem-solving
  - Six Thinking Hats®
- Retrospectives
- Recognition and rewards
- Individual and team assessments
- Data analysis
  - SWOT analysis
- Meetings
- Emotional intelligence
- Organizational cultural intelligence
- Leadership
  - Distributed management and leadership
  - Centralized management and leadership
  - Servant leadership
- Tuckman ladder
- Project management information system
- Virtual collaboration tools
- Etc.

OUTPUTS
- Team performance assessments
- Change requests
- Project management plan updates
  - Resource management plan
  - Schedule baseline
  - Cost baseline
- Project document updates
  - Issue log
  - Lessons learned register
  - Project schedule
  - Resource calendars
  - Team charter
  - Project team assignments
- Enterprise environmental factor updates
- Organizational process asset updates
- Etc.

1.4 GOVERNANCE — Manage Project Execution

INPUTS
- Project management plan
  - Any component
- Project documents
  - Change log
  - Lessons learned register
  - Milestone list
  - Project communications
  - Project schedule
  - Requirements traceability matrix
  - Risk register
  - Risk report
- Approved change requests
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Project management information system
- Meetings
  - Daily coordination meetings
- Etc.

OUTPUTS
- Deliverables
- Work performance data
- Issue log
- Change requests
- Project management plan updates
  - Any component
- Project document updates
  - Activity list
  - Assumption log
  - Lessons learned register
  - Requirements documentation
  - Risk register
  - Stakeholder register
- Organizational process asset updates
- Etc.

1.5 GOVERNANCE — Manage Quality Assurance

INPUTS
- Project management plan
- Project documents
  - All components
- Organizational process assets
  - Policies
  - Procedures
  - Regulations
- Etc.

TOOLS & TECHNIQUES
- Audits
- Checklists
- Data representation
  - Affinity diagrams
  - Cause-and-effect diagrams
  - Flowcharts
- Decision-making
- Problem-solving
- Process improvement
- Etc.

OUTPUTS
- Quality reports
- Change requests
- Project management plan updates
- Project document updates
- Etc.

1.6 GOVERNANCE — Manage Project Knowledge

INPUTS
- Project management plan
  - All components
- Project documents
  - Lessons learned register
  - Project team assignments
  - Resource breakdown structure
  - Stakeholder register
- Deliverables
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Knowledge management
- Information management
- After-action reviews
- In-progress postmortems
- Storytelling
- Retrospective meetings
- Interpersonal and team skills
  - Active listening
  - Facilitation
  - Leadership
  - Networking
  - Political awareness
- Etc.

OUTPUTS
- Lessons learned register
- Project management plan updates
  - Any component
- Organizational process asset updates
- Etc.

7.5 RISK — Implement Risk Responses

INPUTS
- Project management plan
  - Risk management plan
- Project documents
  - Lessons learned register
  - Risk register
  - Risk report
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Interpersonal and team skills
  - Influencing
- Project management information system
- Etc.

OUTPUTS
- Change requests
- Project document updates
  - Issue log
  - Lessons learned register
  - Risk register
  - Risk report
- Etc.

5.4 STAKEHOLDERS — Manage Stakeholder Engagement

INPUTS
- Project management plan
  - Communications management plan
  - Risk management plan
  - Stakeholder engagement plan
  - Change management plan
- Project documents
  - Change log
  - Issue log
  - Lessons learned register
  - Stakeholder register
  - Risk register
  - Status report
  - Project schedule
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Communication skills
  - Feedback
- Interpersonal and team skills
  - Conflict management
  - Cultural awareness
  - Negotiation
  - Observation/—conversation
  - Political awareness
- Ground rules
- Meetings
- Etc.

OUTPUTS
- Change requests
- Project management plan updates
  - Communications management plan
  - Stakeholder engagement plan
- Project document updates
  - Change log
  - Issue log
  - Lessons learned register
  - Stakeholder register
- Etc.

5.5 STAKEHOLDERS — Manage Communications

INPUTS
- Project management plan
  - Resource management plan
  - Communications management plan
  - Stakeholder engagement plan
- Project documents
  - Change log
  - Issue log
  - Lessons learned register
  - Quality report
  - Risk report
  - Stakeholder register
- Work performance reports
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Communication technology
- Communication methods
- Communication skills
  - Communication competence
  - Feedback
  - Nonverbal communication
  - Presentations
- Project management information system
- Project reporting
- Interpersonal and team skills
  - Active listening
  - Conflict management
  - Cultural awareness
  - Meeting management
  - Networking
  - Political awareness
- Meetings
- Etc.

OUTPUTS
- Project communications
- Project management plan updates
  - Communications management plan
  - Stakeholder engagement plan
- Project document updates
  - Issue log
  - Lessons learned register
  - Project schedule
  - Risk register
  - Stakeholder register
- Organizational process asset updates
- Etc.

MONITORING & CONTROLLING

2.5 SCOPE — Monitor and Control Scope

INPUTS
- Project management plan
  - Scope management plan
  - Quality management plan
- Project documents
  - Requirements documentation
  - Performance measurement baseline
  - Quality test and evaluation documents
- Quality metrics
- Approved change requests
- Deliverables
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Data analysis
  - Variance analysis
  - Trend analysis
  - Root cause analysis
- Performance reviews
- Audits and inspections
- Testing/product evaluations
- Data representations
- Process automations
- Etc.

OUTPUTS
- Quality reports
- Verified deliverables
- Change requests
- Quality control measurements
- Project management plan updates
  - Scope management plan
  - Quality management plan
- Project document updates
  - Requirements documentation
  - Lessons learned register
- Work performance information
- Etc.

2.6 SCOPE — Validate Scope

INPUTS
- Project documents
  - Requirements documentation
  - Scope baseline
  - Quality reports
  - Work performance data
  - Quality control measurements
- Verified deliverables
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Data gathering
- Data analysis
- Inspection
- Decision-making
- Customer talks and tests
- Process analysis
- Review meetings
- Etc.

OUTPUTS
- Accepted deliverables
- Change requests
- Project document updates
  - Quality reports
  - Work performance information
  - Requirements documentation
- Lessons learned updates
- Etc.

1.7 GOVERNANCE — Monitor and Control Project Performance

INPUTS
- Project management plan
  - Any component
- Project documents
  - Assumption log
  - Basis of estimates
  - Cost forecasts
  - Issue log
  - Lessons learned register
  - Milestone list
  - Quality reports
  - Risk register
  - Risk report
  - Schedule forecasts
- Work performance information
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data analysis
  - Alternative analysis
  - Cost-benefit analysis
  - Earned value analysis
  - Root cause analysis
  - Trend analysis
  - Variance analysis
- Decision-making
  - Voting
- Meetings
- Project dashboards
- Visual controls
- Information radiators
- Etc.

OUTPUTS
- Work performance reports
- Change requests
- Project management plan updates
  - Any component
- Project document updates
  - Cost forecasts
  - Issue log
  - Lessons learned register
  - Risk register
  - Schedule forecasts
- Etc.

1.8 GOVERNANCE — Assess and Implement Changes

INPUTS
- Project management plan
  - Change management plan
  - Configuration management plan
  - Scope baseline
  - Schedule baseline
  - Cost baseline
- Project documents
  - Basis of estimates
  - Change log
  - Requirements traceability matrix
  - Risk report
- Work performance reports
- Change requests
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Change control tools
- Data analysis
  - Alternative analysis
  - Cost-benefit analysis
- Decision-making
  - Voting
  - Autocratic decision-making
  - Multicriteria decision analysis
- Meetings
- Integrated change control
- Backlog management
- Etc.

OUTPUTS
- Approved change requests
- Project management plan updates
  - Any component
- Project document updates
  - Change log
- Etc.

3.3 SCHEDULE — Monitor and Control Schedule

INPUTS
- Project management plan
  - Schedule management plan
  - Scope baseline
  - Performance measurement baseline
- Product backlog
- Project documents
  - Lessons learned register
  - Project calendars
  - Project schedule
  - Resource calendars
  - Risk register
  - Schedule data
- Work performance data
- Work performance reports
- Etc.

TOOLS & TECHNIQUES
- Data analysis
  - Earned value analysis
  - Burnup/—burndown chart
  - Performance reviews
  - Trend analysis
  - Variance analysis
  - What-if scenario analysis
- Critical path method
- Critical chain method
- Project management information system
- Resource optimization
- Leads and lags
- Schedule compression
- Branch and bound
- Velocity
- Daily coordination meetings
- Sprint reviews
- Backlog refinement
- Etc.

OUTPUTS
- Work performance information
- Schedule forecasts
- Change requests
- Project management plan updates
  - Schedule management plan
  - Schedule baseline
  - Cost baseline
  - Performance measurement baseline
- Project document updates
  - Assumption log
  - Basis of estimates
  - Lessons learned register
  - Project schedule
  - Resource calendars
  - Schedule data
- Etc.

4.4 FINANCE — Monitor and Control Finances

INPUTS
- Project management plan
  - Financial management plan
  - Cost baseline
  - Performance measurement baseline
- Project documents
  - Lessons learned register
  - Project funding requirements
- Work performance data
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data analysis
  - Earned value analysis
  - Trend analysis
  - Reserve analysis
- To-complete performance index
- Project management information system
- Etc.

OUTPUTS
- Work performance information
- Revenue and cost forecasts
- Change requests
- Funding proposals
- Project management plan updates
  - Financial management plan
  - Cost baseline
  - Performance measurement baseline
- Project document updates
  - Assumption log
  - Basis of estimates
  - Cost estimates
  - Lessons learned register
  - Risk register
- Etc.

5.6 STAKEHOLDERS — Monitor Stakeholder Engagement

INPUTS
- Project management plan
  - Resource management plan
  - Communications management plan
  - Stakeholder engagement plan
- Project documents
  - Issue log
  - Lessons learned register
  - Project communications
  - Risk register
  - Stakeholder register
- Work performance data
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Data analysis
  - Alternative analysis
  - Root cause analysis
  - Stakeholder analysis
- Decision-making
  - Multicriteria decision analysis
  - Voting
- Data representation
  - Stakeholder engagement assessment matrix
- Communication skills
  - Feedback
  - Presentations
- Interpersonal and team skills
  - Active listening
  - Cultural awareness
  - Leadership
  - Networking
  - Political awareness
- Meetings
- Etc.

OUTPUTS
- Work performance information
- Change requests
- Project management plan updates
  - Resource management plan
  - Communications management plan
  - Stakeholder engagement plan
- Project document updates
  - Issue log
  - Lessons learned register
  - Risk register
  - Stakeholder register
- Etc.

5.7 STAKEHOLDERS — Monitor Communications

INPUTS
- Project management plan
  - Resource management plan
  - Communications management plan
  - Stakeholder engagement plan
- Project documents
  - Issue log
  - Lessons learned register
  - Project communications
- Work performance data
- Enterprise environmental factors
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Project management information system
- Data representation
  - Stakeholder engagement assessment matrix
- Interpersonal and team skills
  - Observation/—conversation
- Meetings
- Etc.

OUTPUTS
- Work performance information
- Change requests
- Project management plan updates
  - Communications management plan
  - Stakeholder engagement plan
- Project document updates
  - Issue log
  - Lessons learned register
  - Stakeholder register
- Etc.

7.6 RISK — Monitor Risks

INPUTS
- Project management plan
  - Risk management plan
- Project documents
  - Issue log
  - Lessons learned register
  - Risk register
  - Risk report
- Work performance data
- Work performance reports
- Etc.

TOOLS & TECHNIQUES
- Data analysis
  - Technical performance analysis
  - Reserve analysis
- Audits
- Meetings
- Etc.

OUTPUTS
- Work performance information
- Change requests
- Project management plan updates
  - Any component
- Project document updates
  - Assumption log
  - Issue log
  - Lessons learned register
  - Risk register
  - Risk report
- Organizational process asset updates
- Etc.

6.5 RESOURCES — Monitor and Control Resourcing

INPUTS
- Project management plan
  - Resource management plan
- Project documents
  - Issue log
  - Lessons learned register
  - Physical or virtual resource assignments
  - Project schedule
  - Resource breakdown structure
  - Risk register
- Work performance data
- Agreements
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Data analysis
  - Alternative analysis
  - Cost-benefit analysis
  - Performance reviews
  - Trend analysis
- Problem-solving
- Interpersonal and team skills
  - Negotiation
  - Influencing
- Project management information system
- Value stream mapping
- Continuous improvement
- Theory of constraints
- Control charts
- Branch and bound
- Etc.

OUTPUTS
- Work performance information
- Change requests
- Project management plan updates
  - Resource management plan
  - Schedule baseline
  - Cost baseline
- Project document updates
  - Assumption log
  - Issue log
  - Lessons learned register
  - Physical resource assignments
  - Resource breakdown structure
  - Risk register
- Etc.

CLOSING

1.9 GOVERNANCE — Close Project or Phase

INPUTS
- Project charter
- Project management plan
  - All components
- Project documents
  - Assumption log
  - Basis of estimates
  - Change log
  - Issue log
  - Lessons learned register
  - Milestone list
  - Project communications
  - Quality reports
  - Requirements documentation
  - Risk register
  - Risk report
- Accepted deliverables
- Business documents
  - Business case
  - Benefits management plan
- Agreements
- Procurement documentation
- Organizational process assets
- Etc.

TOOLS & TECHNIQUES
- Expert judgment
- Data analysis
  - Document analysis
  - Regression analysis
  - Trend analysis
  - Variance analysis
- Meetings
- Etc.

OUTPUTS
- Project document updates
  - Lessons learned register
- Final product, service, or result transition
- Final report
- Organizational process asset updates
- Etc.

PMBOK® PERFORMANCE DOMAINS
AND THE 40 PROJECT MANAGEMENT PROCESSES

Copyright: ©2025 Project Management Institute, Inc. All rights reserved.

Explanatory Note: This flow ***does not replace*** the need for reading the PMBOK® Guide. The PMBOK® Guide contains a deep explanation of all processes including Inputs, Tools & Techniques and Outputs that are listed in this flow.

Adaptation: Ricardo Viana Vargas **Design:** Sergio Alves Lima Jardim

Download this processes flow at

RICARDO-VARGAS.COM

Become a member of the Project Management Institute (PMI) and download the PMBOK® Guide and other standards at:

PMI.org

GOVERNANCE | SCOPE | SCHEDULE | FINANCE | STAKEHOLDERS | RESOURCES | RISK